

Curriculum Vitæ

Benjamin C. Buchmuller

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My research asks why only a few histone mutations drive cancer.

ACADEMIC CAREER

2025/01 – present **Associate Research Scholar**
Princeton University, Department of Chemistry, Princeton, NJ, United States

2022/01 – 2024/12 **Postdoctoral Researcher in Chemical Biology and Epigenetics**
Princeton University, Department of Chemistry, Princeton, NJ, United States
advisor Prof. Dr. Tom W. Muir

EDUCATION

2017/12 – 2021/10 **Doctoral Studies (Dr. rer. nat.) in Chemical Biology (summa cum laude)**
TU Dortmund University, Faculty of Chemistry and Chemical Biology
In collaboration with the Max Planck Institute of Molecular Physiology, Dortmund

2014/10 – 2016/12 **Master of Science (M. Sc.) in Molecular Biosciences**
Heidelberg University, Faculty of Biosciences
In collaboration with the German Cancer Research Center (DKFZ), Heidelberg

2011/10 – 2014/09 **Bachelor of Science (B. Sc.) in Life Science**
University of Konstanz, Department of Chemistry

FELLOWSHIPS AND AWARDS

2022/01 – 2024/12 **EMBO Postdoctoral Fellow**
European Molecular Biology Organization, Heidelberg, Germany

2023/11 **Biomedizin-Preis 2023, Kategorie: Nachwuchs (Young Talent Award)**
Verein zur Förderung der biomedizinischen und klinischen Forschung, Dortmund, Germany

2022/07 **Rainer Rudolph Prize**
Rainer-Rudolph-Stiftung, Halle, Germany

2019/11 – 2022/02 **Add-on Fellowship for Interdisciplinary Life Science**
Joachim Herz Stiftung, Hamburg, Germany

2019/06 **CSHL Helmsley Fellow**
The Leona M. and Harry B. Helmsley Charitable Trust, New York, NY, United States

2017/12 – 2021/10 **Fellow of the International Max Planck Research School for Living Matter**
Max Planck Institute of Molecular Physiology, Dortmund, Germany

PROFESSIONAL AFFILIATIONS AND SERVICES

2026/02 – present **Review Commons (EMBO Press, ASAPbio)**, Reviewer

2025/06 – present **BMC Biotechnology (BioMed Central)**, Reviewer (ad hoc)

2025/06 – present **Functional & Integrative Genomics (Springer)**, Reviewer (ad hoc)

2023/10 – present **Journal of Molecular Evolution (Springer)**, Reviewer (ad hoc)

2023/01 – present **American Chemical Society**, Member

RESEARCH EXPERIENCE

- 2017/12 – 2021/11 **PhD Student in Chemical Biology of Nucleic Acids**
TU Dortmund University, Faculty of Chemistry and Chemical Biology, Dortmund, Germany
Re-engineered proteins for the selective recognition of epigenetic DNA modifications
advisor Prof. Dr. Daniel Summerer
committee Prof. Dr. Dr. h. c. Herbert Waldmann, Prof. Dr. Andrea Musacchio
- 2017/01 – 2017/11 **Research Scientist in Functional Genomics**
Center for Molecular Biology at Heidelberg University (ZMBH), Heidelberg, Germany
referee Prof. Dr. Michael Knop
- 2016/05 – 2016/12 **Master's Thesis in Synthetic Biology and Functional Genomics**
Center for Molecular Biology at Heidelberg University (ZMBH), Heidelberg, Germany
Developed a method to create proteome-wide yeast clone libraries of in-frame tagged proteins using CRISPR/Cas12a and synthetic oligonucleotide pools
supervisor Prof. Dr. Michael Knop
- 2015/10 – 2016/01 **Intern Student in Research & Development**
Bicycle Therapeutics Ltd., Cambridge, United Kingdom
Large-scale peptide synthesis, work-up, post-synthetic cyclization and quality control
referee Dr. Daniel Teufel
- 2014/01 – 2014/09 **Bachelor's Thesis in Molecular Cell Biology**
University of Konstanz, Department of Chemistry, Konstanz, Germany
Synthesized a photo-affinity labeled small-molecule inhibitor of cell adhesion and knocked-out putative target proteins using CRISPR/Cas9
supervisors Prof. Dr. Christof Hauck (biology) and Dr. Thomas Böttcher (chemistry)

TEACHING AND MENTORING

- 2023/01 – 2023/08 **Princeton University, Department of Chemistry**
– trained and independently supervised 2 graduate students 2023, 2025
- 2017/12 – 2021/11 **TU Dortmund University, Faculty of Chemistry and Chemical Biology**
– trained and independently supervised 2 Master of Science students 2018, 2019
– trained and independently supervised 3 Bachelor of Science students 2018, 2019, 2020
– hosted 1 summer student 2018
– teaching fellow practical course 'Biochemistry 1' Summer 2018, 2020
– teaching fellow practical course 'Advanced Methods in Protein Modification' Winter 2018
- 2018/01 – present **DKFZ Heidelberg, Heidelberger Life-Science Lab**
– instructor for high-school students in 'R Programming' 2021
– instructor in upgrade training for high-school teachers 'Genome Editing' 2018
- 2017/01 – 2017/11 **Heidelberg University, Faculty of Biosciences**
– trained 1 ERASMUS+ (European exchange program) Master student 2017
– hosted 1 summer student 2017
– teaching fellow practical course 'Quantitative Cell Biology' Winter 2017

ORIGINAL RESEARCH

- [21] L. Engelhard*, D. Schiller*, M. S. Zambrano-Mila, K. Keliuotyte, B. Buchmuller, S. Tiwari, J. Imig, A. Simeone, C. Schröter, S. Becker, D. Summerer. HM-DyadCap – Capture and Mapping of 5-Hydroxymethylcytosine/5-Methylcytosine CpG Dyads in Mammalian DNA. *Nucleic Acids Res.* **2026**, DOI: 10.1101/nar/gkag389
- [20] B. Kosel, K. Bigler, B. C. Buchmuller, S. R. Acharyya, R. Linser, D. Summerer. Evolved readers of 5-carboxylcytosine CpG dyads reveal a high versatility of the methyl-CpG-Binding domain for recognition of noncanonical epigenetic marks. *Angew. Chem. Int. Ed.* **2024**, DOI: 10.1002/anie.202318837.
- [19] H. Singh, C. K. Das, B. C. Buchmuller, L. V. Schäfer, D. Summerer, R. Linser. Epigenetic CpG duplex marks probed by an evolved DNA reader via a well-tempered conformational plasticity. *Nucl. Acids Res.* **2023**, 51 (12), 6495–6506, DOI: 10.1093/nar/gkad134.
- [18] J. Nowacki, M. Malenica, S. Schmeing, D. Schiller, B. Buchmuller, G. Amrahova, P. 't Hart. A translational repression reporter assay for the analysis of RNA-binding protein consensus sites. *RNA Biol.* **2023**, 20(1), 85–94, DOI: 10.1080/15476286.2023.2192553.
- [17] S. Palei, J. Weisner, M. Vogt, R. Gontla, B. Buchmuller, C. Ehrh, T. Grabe, S. Kleinbölting, M. Müller, G. H. Clever, D. Rauh, D. Summerer. A high-throughput effector screen identifies a novel small molecule scaffold for inhibition of ten-eleven translocation dioxygenase 2. *RSC Med. Chem.* **2022**, 13, 1540–1548, DOI: 10.1039/d2md00186a.
- [16] A. Jung, Á. Muñoz-López, B. C. Buchmuller, S. Banerjee, D. Summerer. Imaging-based in situ analysis of 5-methylcytosine at low repetitive single gene loci with transcription-activator-like effector probes. *ACS Chem. Biol.* **2022**, 18(2), 230–236, DOI: 10.1021/acscchembio.2c00857.
- [15] B. C. Buchmuller, J. Dröden, H. Singh, S. Palei, M. Drescher, R. Linser, D. Summerer. Evolved DNA duplex readers for strand-asymmetrically modified 5-hydroxymethylcytosine/5-methylcytosine CpG dyads. *J. Am. Chem. Soc.* **2022**, 144(7), 2987–2993, DOI: 10.1021/jacs.1c10678.
- [14] Á. Muñoz-López, A. Jung, B. Buchmuller, J. Wolffgramm, S. Maurer, A. Witte, D. Summerer. Engineered TALE repeats for enhanced imaging-based analysis of cellular 5-methylcytosine. *ChemBioChem* **2020**, 22(4), 645–651, DOI: 10.1002/cbic.202000563.
- [13] J. Wolffgramm, B. Buchmuller, S. Palei, Á. Muñoz-López, J. Kanne, P. Janning, M. Schweiger, D. Summerer. Light-activation of DNA-methyltransferases. *Angew. Chem. Int. Ed.* **2021**, 60, 13507, DOI: 10.1002/anie.202103945.
- [12] S. Palei, B. Buchmuller, J. Wolffgramm, Á. Muñoz-López, S. Jung, P. Czodrowski, D. Summerer. Light-activatable TET dioxygenases reveal dynamics of 5-methylcytosine oxidation and transcriptome reorganization. *J. Am. Chem. Soc.* **2020**, 142(16), 7289–7294, DOI: 10.1021/jacs.0c01193.
- [11] J. Fueller, K. Herbst, M. Meurer, K. Gubicza, B. Kurtulmus, J. D. Knopf, D. Kirrmaier, B. C. Buchmuller, G. Pereira, M. K. Lemberg, M. Knop. CRISPR–Cas12a-assisted PCR tagging of mammalian genes. *J. Cell Biol.* **2020**, 219(6), 615–624, DOI: 10.1083/jcb.201910210.
- [10] Á. Muñoz-López, B. Buchmuller, J. Wolffgramm, A. Jung, M. Hussong, J. Kanne, M. R. Schweiger, D. Summerer. Designer receptors for nucleotide-resolution analysis of genomic 5-methylcytosine by cellular imaging. *Angew. Chem. Int. Ed.* **2020**, 59(23), 8927–8931, DOI: 10.1002/anie.202001935.
- [9] B. C. Buchmuller, B. Kosel, D. Summerer. Complete profiling of methyl-CpG-binding domains for combinations of cytosine modifications at CpG dinucleotides reveals differential read-out in normal and Rett-associated states. *Sci. Rep.* **2020**, 10, 4053, DOI: 10.1038/s41598-020-61030-1.
- [8] B. C. Buchmuller*, K. Herbst*, M. Meurer, D. Kirrmaier, E. Sass, E. D. Levy, M. Knop. Pooled clone collections by multiplexed CRISPR–Cas12a-assisted gene tagging in yeast. *Nat. Commun.* **2019**, 10, 2960, DOI: 10.1038/s41467-019-10816-7.
- [7] M. Gieß, Á. Muñoz-López, B. Buchmuller, G. Kubik, D. Summerer. Programmable protein-DNA cross-linking for the direct capture and quantification of 5-formylcytosine. *J. Am. Chem. Soc.* **2019**, 141(24), 9453–9457, DOI: 10.1021/jacs.9b01432.

- [6] S. Maurer, B. Buchmuller, C. Ehrh, J. Jasper, O. Koch, D. Summerer. Overcoming conservation in TALE–DNA interactions: A minimal repeat scaffold enables selective recognition of oxidized 5-methylcytosine. *Chem. Sci.* **2018**, 9(36), 7247–7252, DOI: 10.1039/c8sc01958d.
- [5] M. Meurer, Y. Duan, E. Sass, I. Kats, K. Herbst, B. C. Buchmuller, V. Dederer, F. Huber, D. Kirrmaier, M. Štefl, K. Van Lear, T. P. Dick, M. K. Lemberg, A. Khmelinskii, E. D. Levy, M. Knop. Genome-wide C-SWAT library for high-throughput yeast genome tagging. *Nat. Methods* **2018**, 15 (8), 598–600, DOI: 10.1038/s41592-018-0045-8.
- [4] C. Liolios, B. Buchmuller, U. Bauder-Wüst, M. Schäfer, K. Leotta, U. Haberkorn, M. Eder, K. Kopka. Monomeric and dimeric [68-Ga]-labeled bombesin analogues for PET imaging of tumors expressing gastrin-releasing peptide receptors. *J. Med. Chem.* **2018**, 61 (5), 2062–2074, DOI: 10.1021/acs.jmedchem.7b01856.

REVIEWS, BOOK CHAPTERS, MONOGRAPHS

- [3] B. Buchmuller. Deciphering strand-asymmetrically modified CpG dyads in the DNA double-helix. An evolutionary approach. Dissertation, TU Dortmund University, Dortmund, Germany, **2021**, DOI: 10.17877/DE290R-22422.
- [2] B. Buchmuller, Á. Muñoz-López, M. Gieß, D. Summerer. Design and application of DNA modification-specific transcription-activator-like effectors. In *DNA Modifications. Methods Mol. Biol.*; A. Ruzov, M. Gering (Eds.); Humana: New York, NY, **2021**; 2198, DOI: 10.1007/978-1-0716-0876-0_29.
- [1] B. Buchmuller, A. Jung, Á. Muñoz-López, D. Summerer. Programmable tools for the targeted analysis of epigenetic DNA modifications. *Curr. Opin. Chem. Biol.* **2021**, 63, 1–10, DOI: 10.1016/j.cbpa.2021.01.002.

PATENTS

- B. Buchmuller; D. Summerer (Technische Universität Dortmund). Method for determining 5-methylcytosine configurations in DNA. PCT Pat. Appl. WO 19 220 082.2, 2020.

POSTER AND CONFERENCE PRESENTATIONS

- B. Buchmuller. *Reading complex chemical modifications in DNA*. Invited talk delivered at the 7th Halle Conference on Recombinant Proteins, Halle (Saale), Germany, October 2022.
- B. Buchmuller. *An evolved DNA double-strand reader of strand-asymmetric 5-methylcytosine oxidation at CpG dinucleotides*. Invited talk delivered at the 3rd Annual (Virtual) Symposium of the International Chemical Biology Society (ICBS) Students' Chapter Dortmund, Dortmund, Germany, October 2021.
- B. Buchmuller, D. Summerer. *A DNA double-strand reader of 5-methylcytosine oxidation at CpG dinucleotides*. Poster presentation delivered at the EMBL Virtual Conference Chromatin and Epigenetics, European Molecular Biology Laboratory (EMBL), Heidelberg, Germany, May 2021.
- B. Buchmuller, D. Summerer. *A set of engineered DNA-binding proteins to explore 5-methylcytosine oxidation across the DNA double-strand*. Poster presentation delivered at the 13. Tag der Chemie, Technische Universität Dortmund, Dortmund, Germany, February 2020.
- B. Buchmuller, M. Meurer, E. Sass, K. Herbst, D. Kirrmaier, E. Levy, M. Knop. *Generation of pooled genome-wide yeast clone libraries by CRISPR/Cpf1-assisted tag integration*. Poster presentation delivered at the 28th International Conference on Yeast Genetics and Molecular Biology, Prague, Czech Republic, August 2017.
- B. Buchmuller, C. Liolios, M. Schäfer, U. Bauder-Wüst, M. Eder, U. Haberkorn, K. Kopka. *[68-Ga]-labeled bombesin tracers for PET imaging of prostate cancer*. Invited talk delivered at the 54. Jahrestagung der Deutschen Gesellschaft für Nuklearmedizin, Dresden, Germany, April 2016.